

- 6 (a) The charge is stationary.
The charge is moving parallel to the direction of the magnetic field.

(b) (i) $v_{\perp} = 4.7 \times 10^5 \times \sin 20^\circ = 1.61 \times 10^5 \text{ m s}^{-1}$

- (ii) Magnetic force provides centripetal force

$$Bqv_{\perp} = \frac{m(v_{\perp})^2}{r}$$

$$0.12 \times 3.2 \times 10^{-19} = \frac{m \times 1.6075 \times 10^5}{\frac{1}{2} \times 0.056}$$

$$m = 6.69 \times 10^{-27} \text{ kg}$$

(iii) $T = \frac{2\pi r}{v_{\perp}} = \frac{\pi \times 0.056}{1.6075 \times 10^5} = 1.0944 \times 10^{-6} \text{ s}$

$$\begin{aligned}
 x &= v \cos 20^\circ \times T \\
 &= 4.7 \times 10^5 \times \cos 20^\circ \times 1.0944 \times 10^{-6} \\
 &= 0.483 \text{ m}
 \end{aligned}$$

- (iv) Electrons will travel in a helical path with smaller radius.
 Electrons will travel in a helical path with smaller period.
 Electrons will travel in a helical path with smaller x.
Direction of rotation of the helical path of the electrons will be opposite to that of the particles.

Additional Information:

$$\begin{aligned}
 Bqv_{\perp} &= \frac{m(v_{\perp})^2}{r} \\
 r &= \frac{mv_{\perp}}{Bq}
 \end{aligned}$$

Mass of electron is 9.11×10^{-31} kg and charge is -1.60×10^{-19} C. Hence, r_e will be much smaller.

Time $\left(T = \frac{2\pi r}{v_{\perp}} \right)$ taken for the electrons to complete one revolution in the helix will also be much smaller and x will be smaller.